

```
static char c;

static ssize_t my_read(struct file *f, char __user *buf, size_t
len, loff_t *off)
{
    printk(KERN_INFO "Driver: read()\n");
    if (copy_to_user(buf, &c, 1) != 0)
        return -EFAULT;
    else
        return 1;
}

static ssize_t my_write(struct file *f, const char __user *buf,
size_t len, loff_t *off)
{
    printk(KERN_INFO "Driver: write()\n");
    if (copy_from_user(&c, buf + len - 1, 1) != 0)
        return -EFAULT;
    else
        return len;
}
```

Then Shweta repeated the usual build-and-test steps as follows:

- Build the modified 'null' driver (.ko file) by running *make*.

- Load the driver using *insmod*.
- Write into */dev/mynull*, say, using *echo -n "Pugs" > /dev/mynull*
- Read from */dev/mynull* using *cat /dev/mynull* (stop by using *Ctrl+C*)
- Unload the driver using *rmmod*.

On *cat'ing /dev/mynull*, the output was a non-stop infinite sequence of 's', as *my_read()* gives the last one character forever. So, Pugs intervened and pressed *Ctrl+C* to stop the infinite read, and tried to explain, "If this is to be changed to 'the last character only once', *my_read()* needs to return 1 the first time, and zero from the second time onwards. This can be achieved using *off* (the fourth parameter of *my_read()*)." Shweta nodded her head obligingly, just to bolster Pugs' ego.  **END**

By: Anil Kumar Pugalia

The author is a freelance trainer in Linux internals, Linux device drivers, embedded Linux and related topics. Prior to this, he was at Intel and Nvidia. He has been working with Linux since 1994. A gold medalist from the Indian Institute of Science, Linux and knowledge sharing are two of his many passions. More about him can be found at <http://profession.sanka-pugs.com>, including information about his various Linux related training sessions and workshops. He can be reached at email@sanka-pugs.com.

www.mafiree.com

Expert MySQL Design and Architecture
Server Performance Tuning and Query Tuning
High Availability and High Scalability designs
Proactive Monitoring
24 x 7 Support for MySQL servers
Unlimited Tickets

Reduce Operational Cost of Managing servers. Try our Free MySQL audit
www.mafiree.com/free_mysql_audit

 Mafiree

Unit 303, III Floor, Vanguard Rise, 5th Main, Konena Agrahara
Old Airport Road, Bangalore - 560 017 India.
Telephone: +91-80-41155993, Email ID: support@mafiree.com
Website: www.mafiree.com